Supersymmetric sigma-models

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1 $N = 2$ Supersymmetric sigma-models

1.1 de-Rham complex, spin connection

Let M be a $D-$dimensional riemanian manifold with a Levi-Civita connection. At every point of M one can construct an orthonormal basis e_A. Orthonormality condition

$$g(e_A, e_B) = \delta_{AB}. \tag{1}$$

In other words, e_A is a section of a $O(D)$-bundle formed by a set of orthonormal bases at every point of M. We can define a connection on this bundle:

$$\omega_{AB,M} = e_{AN}\Big(\partial_M e_B^N + \Gamma^N_{MP} e_B\Big). \tag{2}$$

Besides, there is an operation on M called the Hodge Star:

$$\begin{aligned} &\star : \Omega^p(M) \to \Omega^{d-p}(M), \\ &\star\,\omega = \frac{1}{(d-p)!}\sqrt{g}\epsilon^{i_1\dots i_p}_{i_{p+1}\dots i_d}\omega_{i_1\dots i_p} dx^{i_{p+1}} \wedge \dots \wedge dx^{i_d}. \end{aligned} \tag{3}$$

There is an inner product on the set of differential forms on M:

$$(\alpha,\beta) = \int_M \alpha \wedge \star\beta = p! \int_M \sqrt{g}\; dx\; \alpha_{i_1\dots i_p}\overline{\beta^{i_1\dots i_p}}, \tag{4}$$

which turns it into a Hilbert space called the de-Rham complex. We note that the definition (4) makes sense only for forms of similar degree. For forms of different degree we set it to be 0 by definition. There following operators act on the de-Rham complex: de-Rham operator d, its conjugate $d^\dagger$ and the Laplace-Beltrami operator $\Delta = -(dd^\dagger + d^\dagger d)$.
The quotient

$$H^p_{DR} = Ker\; d^p / Im\; d^{p-1}, \tag{5}$$

where $d^p : \Omega^p(M) \to \Omega^{p-1}(M)$ is the de-Rham operator acting on p-forms is called the de-Rham cohomology class of M. The numbers $b^p = dim\; H^p_{DR}$ are called the Betti numbers of M. We notice that ω is harmonic iff $d\omega = 0$ и $d^\dagger\omega = 0$, from where it follows clearly that the dimension of the space of harmonic p-forms is equal to b^p. To prove the former, one can do a simple computation:

$$(\omega, \Delta\omega) = (\omega, (d^\dagger d + dd^\dagger)\omega) = (d\omega, d\omega) + (d^\dagger\omega, d^\dagger\omega) \geq 0. \tag{6}$$

Our goal is to determine some geometric properties of M by studying SQM on M. By considering a sigma-model with $N = 2$ supercharges, we will establish a bijection between the action of d and $d^\dagger$ on the de-Rham complex and the action of the supercharges on the Hilbert space of our QM system.

1.1.1 Constructing the lagrangian

We begin with the general sigma-model

$$L = \frac{1}{2} g_{MN}\dot{x}^M\dot{x}^N. \tag{7}$$

Now we make it supersymmetric by adding D real superfields:

$$X^M = x^M + \theta\psi^M + \bar{\psi}^M\bar{\theta} + \theta\bar{\theta}F^M, \tag{8}$$

and change the regular derivatives to "covariant":

$$\begin{cases} D = \frac{\partial}{\partial\theta} - i\bar{\theta}\frac{\partial}{\partial t} \\ \bar{D} = -\frac{\partial}{\partial\bar{\theta}} + i\theta\frac{\partial}{\partial t} \end{cases} \tag{9}$$

The resulting action of the SUSY sigma-model:

$$S = \frac{1}{2}\int dt\, d\bar{\theta}\, d\theta g_{MN}\bar{D}X^M DX^N \tag{10}$$

Now let us write this more explicilty:

$$\bar{D}X^M = \bar{\psi}^M + \theta F^M + i\theta\dot{x}^M - i\theta\bar{\theta}\dot{\bar{\psi}}^M, \tag{11}$$

$$DX^N = \psi^N + \bar{\theta}F^N - i\bar{\theta}\dot{x}^N + i\theta\bar{\theta}\dot{\psi}^N \tag{12}$$

$$g_{MN}(X) = g_{MN}(x) + \theta\partial_P g_{MN}(x)\psi^P - \bar{\theta}\partial_P g_{MN}\bar{\psi}^P + \theta\bar{\theta}\Big(\partial_P g_{MN}(x)F^P + \frac{1}{2}\partial_{QP}g_{MN}(\psi^Q\bar{\psi}^P - \bar{\psi}^Q\psi^P)\Big). \tag{13}$$

By singling out the terms of highest order by super-times we get

$$\begin{aligned} L = \frac{1}{2}\Big(g_{MN}(x)\dot{x}^M\dot{x}^N + i\bar{\psi}^M(g_{MN}\dot{\psi}^N + \partial_P g_{MN}\dot{x}^N\psi^P) - i(g_{MN}\dot{\bar{\psi}}^M + \partial_P g_{MN}\dot{x}^M\bar{\psi}^P)\psi^N\Big) + \\ + \frac{1}{2}g_{MN}F^M F^N + \frac{1}{2}\partial_P g_{MN}\Big(\psi^P\bar{\psi}^M F^N - \bar{\psi}^P\psi^N F^M + \bar{\psi}^M\psi^N F^P\Big) + \frac{1}{2}(\partial_{QP}g_{MN})\bar{\psi}^P\bar{\psi}^M\psi^Q\psi^N \end{aligned} \tag{14}$$

This can be simplified further by using the contractions with the antisymmetric combinations of ψ.

$$L = \frac{1}{2}g_{MN}(x)\Big(\dot{x}^M\dot{x}^N + F^M F^N + i(\bar{\psi}^M\nabla\psi^N - \nabla\bar{\psi}^M\psi^N\Big) + \Gamma_{M,PQ}F^M\psi^P\bar{\psi}^Q + \frac{1}{2}(\partial_{PQ}g_{MN})\bar{\psi}^P\bar{\psi}^M\psi^Q\psi^N, \tag{15}$$

where

$$\nabla\psi^N = \dot{\psi}^N + \Gamma^M_{PS}\dot{x}^P\psi^S, \tag{16}$$

We also note that, F^M is not a dynamical variable (since it is of the highest order by X). Therefore we get a constraint

$$\frac{\partial L}{\partial F^M} = 0, \tag{17}$$

from which it is clear that

$$F^M = -\Gamma^M_{PQ}\psi^P\bar{\psi}^Q. \tag{18}$$

After substituting it into the Lagrangian we get

$$L = \frac{1}{2}\Big(\dot{x}^M\dot{x}^N + i(\bar{\psi}^M\nabla\psi^N - \nabla\bar{\psi}^M\psi^N\Big) - \frac{1}{4}R_{PQMN}\bar{\psi}^P\bar{\psi}^M\psi^Q\psi^N, \tag{19}$$

, where we can single out the Riemann tensor:

$$R_{PMQN} = g_{PL}R^L_{QMN} = g_{PL}\Big(\partial_M\Gamma^L_{QN} - \partial_N\Gamma^L_{QM}\Big) + \Gamma_{PMR}\Gamma^R_{QN} - \Gamma_{PQR}\Gamma^R_{MN} \tag{20}$$

For the quantisation we have to switch to the hamiltonian picture, so let us compute the canonical momentum as usual

$$\Pi_M = \frac{\partial L}{\partial\dot{x}^M} = \dot{x}^M + \frac{i}{2}\Big(\partial_P g_{MN} - \partial_N g_{MP}\Big)\psi^N\bar{\psi}^P \tag{21}$$

Supersymmetry transformations

$$\begin{cases} \theta \mapsto \theta + \epsilon, \\ \bar{\theta} \mapsto \bar{\theta} + \bar{\epsilon}, \\ t \mapsto t + i(\epsilon\bar{\theta} - \theta\bar{\epsilon}). \end{cases} \tag{22}$$

induce the following transformations of the fields (keeping (13) in mind):

$$\begin{cases} \delta x^M = \epsilon\psi^M - \bar{\theta}\bar{\psi}^M, \\ \delta\psi^M = \bar{\epsilon}\Big(F^M - i\dot{x}^M\Big), \\ \delta\bar{\psi}^M = \epsilon\Big(i\dot{x}^M - \Gamma^M_{PQ}\psi^P\bar{\psi}^Q\Big). \end{cases} \tag{23}$$

From the Noether theorem we get the classical supercharges

$$\begin{cases} Q = \frac{1}{\sqrt{2}}\psi^M\Big(\Pi_M - \frac{i}{2}\partial_M g_{NP}\psi^N\bar{\psi}^P\Big) \\ \bar{Q} = \frac{1}{\sqrt{2}}\bar{\psi}^M\Big(\Pi_M + \frac{i}{2}\partial_M g_{NP}\psi^N\bar{\psi}^P\Big) \end{cases} \tag{24}$$

We will not go through that derivaton, but we'll show that the Poisson brackets of Q and $\bar{Q}$ with the dynamical variables correspond to(18) with one example:

$$\{\bar{Q}, \psi_M\} = \frac{\partial\bar{Q}}{\partial\bar{\psi}^M} = \Pi_M + \frac{i}{2}\partial_M g_{NP}\psi^N\bar{\psi} - \bar{\psi}^A\frac{\partial\Pi_A}{\partial\bar{\psi}^M} + \frac{i}{2}\bar{\psi}^A\partial_A g_{MN}\psi^N = \dot{x}_M + i\Gamma_{MPQ}\psi^P\bar{\psi}^Q \tag{25}$$

1.1.2 Quantisation

Now our goal is to quantize this system. Remember that for hamiltonian systems with grassmanian variables the Poisson bracket is defined as

$$\{f,g\} = \left(\frac{\partial f}{\partial P_j}\frac{\partial g}{\partial Q_j} - \frac{\partial f}{\partial Q_j}\frac{\partial g}{\partial P_j} - i\Big(\frac{\partial f}{\partial \psi_\alpha}\frac{\partial g}{\partial \bar\psi_\alpha} + \frac{\partial f}{\partial \bar\psi_\alpha}\frac{\partial g}{\partial \psi_\alpha}\Big)\right). \tag{26}$$

It is inconvenient to quantise this systems in the current coordinates, since ψ^M and $\bar\psi^M$ are not canonically conjugate, their Poisson bracket is

$$\{\psi^N, \bar\psi^M\} = -ig^{MN}, \tag{27}$$

therefore we cannot apply the usual quantisation method directly. We can get rid of that obstacle by re-defining the variables

$$\psi^M = \psi_C e^M_C (\text{добавить, что такое } e^m_c). \tag{28}$$

Conjugated variables P_M are obtained by varying the lagrangian with respect to $\dot x^M$ with fixed vielbein components ψ_C. The difference now is that $\dot\psi^M = \frac{d}{dt}(\psi_C e^M_C)$ becomes dependent on $\dot x^M$.

$$P_M = \Pi_M + \frac{\partial \dot\psi^N}{\partial \dot x^M}\frac{\partial L}{\partial \dot\psi^M} + \frac{\partial \dot\psi^N}{\partial \dot x^M}\frac{\partial L}{\partial \dot{\bar\psi}^M} = \Pi_M + \frac{i}{2}\Big((\partial_M e^B_N)e_{AN} - (\partial_M e^N_A)e_{BN}\Big)\psi_A\bar\psi_B. \tag{29}$$

In the new variables the Poisson brackets are just right:

$$\{P_M, x^M\} = \delta^M_N, \ \ \{\psi_A, \bar\psi_B\} = i\delta_{AB}. \tag{30}$$

Supercharges become:

$$\begin{aligned} Q &= \frac{1}{\sqrt{2}}\psi_C e^M_C\Big(P_M - i\omega_{AB,M}\psi_A\bar\psi_B\Big),\\ \bar Q &= \frac{1}{\sqrt{2}}\bar\psi_C e^M_C\Big(P_M - i\omega_{AB,M}\psi_A\bar\psi_B\Big). \end{aligned} \tag{31}$$

Finally, we can quantize:

$$\begin{cases} \hat P_M = -i\partial_M \\ \hat{\bar\psi}_A = \frac{\partial}{\partial\psi_A} \end{cases} \tag{32}$$

We use the normal ordering here

$$\begin{aligned} &e^M_C P_M \mapsto e^M_C \hat P_M \\ &\psi_C\psi_A\bar\psi_B \mapsto \psi_C\psi_A\frac{\partial}{\partial\psi_B} \end{aligned} \tag{33}$$

We get the supercharges:

$$\begin{cases} \hat Q = -\frac{i}{\sqrt 2}\psi^M\Big(\frac{\partial}{\partial x^M} + \omega_{AB,M}\psi_A\frac{\partial}{\partial\psi_B}\Big) \\ \hat{\bar Q} = -\frac{i}{\sqrt 2}e^M_C\frac{\partial}{\partial\psi_C}\Big(\frac{\partial}{\partial x^M} + \omega_{AB,M}\psi_A\frac{\partial}{\partial\psi_B}\Big) \end{cases} \tag{34}$$

that act on the Hilbert space consisting of the wave functions of the form:

$$\Psi(x^M, \psi_A) = A_{M_1\dots M_p}(x)\psi^{M_1}\dots\psi^{M_p}, \tag{35}$$

with skew-symmetric $A_{M_1\dots M_P}$. The inner product is defined with a measure:

$$d\mu = \prod_{I=1}^{D} dx^I\sqrt{g}\prod_{A=1} d\psi_A d\bar\psi_A e^{-\psi_A\bar\psi_A} \tag{36}$$

Let us verify that the inner product of the states with different fermionic charge is zero. Let

$$\begin{aligned} &\Psi_1 = A_{M_1\dots M_p}(x)\psi^{M_1}\dots\psi^{M_p},\\ &\Psi_2 = B_{N_1\dots N_q}(x)\psi^{N_1}\dots\psi^{N_q},\\ &\langle\Psi_1|\Psi_2\rangle = \int_M \sqrt{g}A_{M_1\dots M_p}(x)\overline{B_{N_1\dots N_q}(x)}\int d\psi_A\, d\bar\psi_A e^{-\psi_A\bar\psi_A}\psi_{M_1}\dots\psi_{M_p}\bar\psi_{N_1}\dots\bar\psi_{N_q}. \end{aligned} \tag{37}$$

By substituting $\psi_A \mapsto e^{i\alpha}\psi_A$ we get

$$\langle\Psi_1|\Psi_2\rangle = e^{i\alpha(p-q)}\langle\Psi_1|\Psi_2\rangle, \tag{38}$$

from which it follows that $\langle\Psi_1|\Psi_2\rangle \neq 0$ only if $p = q$. For the states with similar fermion charge the inner produt can be computed using Wick's theorem

$$\langle\Psi_1|\Psi_2\rangle = p! \int_M \sqrt{g} F_{A_1\ldots A_p}(x)\overline{G^{A_1\ldots A_p}} \tag{39}$$

Let's find out how the supercharge $\hat{Q}$ acts on the states. Let

$$\Psi_p = A_{M_1\ldots M_p}(x)\psi^{M_1}\ldots\psi^{M_p},\ \psi^M = e^M_A\psi_A. \tag{40}$$

$$\hat{Q}\Psi = -\frac{i}{\sqrt{2}}\Big((\partial_M A_{M_1\ldots M_p})\psi^M\psi^{M_1}\ldots\psi^{M_p} + pA_{M_1\ldots M_p}e_{AN}\Big((\partial_M e^{M_1}_A) + \omega_{AB,M}e^{M_1}_B\Big)\psi^M\psi^N\psi^{M_2}\ldots\psi^{M_p} \tag{41}$$

Using (2) for the spin-connection, the part of the last term can be transformed into:

$$\Gamma^{M_1}_{MN}\psi^M\psi^N. \tag{42}$$

Finally:

$$\hat{Q}\Psi = -\frac{i}{\sqrt{2}}(\partial_M A_{M_1\ldots M_p})\psi^M\psi^{M_1}\ldots\psi^{M_p} \tag{43}$$

This expression is consistent with the action of the de-Rham operator on p-forms $\omega = \omega_{i_1\ldots i_p}dx^{i_1}\wedge\ldots\wedge dx^{i_p}$ from the de-Rham complex:

$$d\omega = \partial_m\omega_{i_1\ldots i_p}dx^m\wedge dx^{i_1}\wedge\ldots\wedge dx^{i_p} \tag{44}$$

Respectively, the de-Rham complex is a Hilbert space with an inner product:

$$(\omega,\beta) = \int_M \omega\wedge\star\beta = \int \alpha_{i_1\ldots i_p}\beta^{i_1..i_p}\sqrt{g}\prod_{I=1}^{M} dx^I \tag{45}$$

Since $d^\dagger$ is adjoint to d with respect to the measure (14), i.e for any forms ω,β

$$(\omega, d\beta) = (d^\dagger\omega,\beta), \tag{46}$$

the supercharge $\hat{Q}^\dagger$, adjoint $\hat{Q}$ corresponds to $d^\dagger$.

1.1.3 Outcomes

The obtained connection added to the prior knowledge of SQM systems allows to make the following conclusions:
1) An arbitrary form ω admits a decomposition:

$$\omega = d\beta + d^\dagger\alpha + h, \tag{47}$$

where h is a harmonic form, $(dd^\dagger + d^\dagger d)h = 0$. We note that from a supersymmetric perspective $\{d,d^\dagger\}$ corresponds to the hamiltonian, therefore a harmonic form corresponds to a state with zero energy, and exact and co-exact forms β and α correspond to the states from fermionic and bosonic sectors. 2) The Euler characteristic of M is equal to the Witten index of our SQM. Clearly, p-cohomology classes correspond to the vacuum states of fermionic charge p, there fore the dimension of H^p_{DR} counts such states. The parity p. Thus, the Euler characteristic $\chi(M) = \sum_p(-1)^p\, b_p(M)$ is precisely the Witten index $I_W = n_B - n_F$.
3) Gauss-Bonnet formula for arbitrary dimension (in future talks)

1.2 Dolbeault complex

Now let M be a $D = 2d-$dimensioanl hermitian manifold with an hermitian metric

$$g = h_{m\bar{n}}dz^m\otimes d\bar{z}^{\bar{n}} + h_{\bar{m}n}dz^{\bar{m}}\otimes dz^n, \tag{48}$$

$h_{m\bar{n}} = \overline{h_{\bar{m}n}}$ is an hermitian matrix. Differential forms are now indexed by two integers (p,q):

$$\omega_{p,q} = \omega_{i_1,..i_p,j_1,\ldots j_q}dz^{i_1}\wedge\ldots\wedge dz^{i_p}\wedge dz^{j_q}\wedge\ldots\wedge dz^{j_q}. \tag{49}$$

Similarly to the real case, the Hodge Star can be defined:

$$\star\omega_{p,q} = \det h\frac{1}{(d-p)!(d-q)!}\varepsilon^{m_1\ldots m_p}_{m_{p}+1\ldots m_d}\varepsilon^{n_1\ldots n_q}_{\bar{n}_{q+1}\ldots\bar{n}_d}\overline{\alpha_{m_1\ldots m_p n_1\ldots n_q}}dz^{m_{p+1}}\wedge\ldots\wedge dz^{m_d}\wedge d\bar{z}^{\bar{n}_{q+1}}\wedge\ldots\wedge d\bar{z}^{\bar{n_d}}. \tag{50}$$

The inner product of forms is defined the same way:

$$(\alpha_{p,q},\beta_{p,q}) = \int_M \alpha_{p,q}\wedge\star\beta_{p,q} = p!q!\int_M deth\; dzd\bar{z}\;\alpha_{r_1\ldots r_p\bar{s}_1\ldots\bar{s}_q}\overline{\beta^{r_1\ldots r_p\bar{s}_1\ldots\bar{s}_q}} \tag{51}$$

On the set of (p,0)-forsm act operators ∂ and its adjoint $\partial^\dagger$. $\{\partial,\partial^\dagger\} = \Delta$ is called the Dolbeault laplacian. The set of (p,0)-forms we will call the Dolbeault complex. Similarly, on the set (0,q)-forms called the anti-Dolbeault complex act operators $\bar{\partial}$ and $\bar{\partial}^\dagger$. We shall note that $\partial + \bar{\partial} = d$, $\partial^\dagger + \bar{\partial}^\dagger = d^\dagger$.

1.2.1 Constructing the Lagrangian

We will construct the sigma model from d holomorphic and d anti-holomorphic superfields

$$\begin{cases} Z^m = z^m + \sqrt{2}\theta\psi^m - i\theta\bar{\theta}\dot{z}^m \\ \bar{Z}^m = \bar{z}^m + \sqrt{2}\bar{\theta}\bar{\psi}^m + i\theta\bar{\theta}\dot{\bar{z}}^m \end{cases} \tag{52}$$

that is invariant under the supersymmetry transformations. Its Lagrangian is:

$$L = \frac{1}{4}\int d\bar{\theta}d\theta h_{m\bar{n}}(\bar{Z}, Z)\bar{D}\bar{Z}^n DZ^m \tag{53}$$

1.2.2 Quantisation

Now we will proceed with the similar actions:
1) Get the classical supercharges from the Noether theorem
2) Change the variables to $\{p_c, p_{\bar{c}}\}$ and $\{\psi^c, \bar{\psi}^{\bar{c}}\}$, by changing the basis in the tangent space.
Omitting the calculations on steps 1-2, we'll proceed with an answer for the classical supercharges in terms of the "right"momentum variables:

$$\begin{aligned} Q &= e^m_c\psi^c\Big(p_m - i\omega_{a\bar{b},m}\psi^a\bar{\psi}^{\bar{b}}\Big), \\ \bar{Q} &= e^{\bar{m}}_c\psi^c\Big(p_{\bar{m}} - i\omega_{a\bar{b},\bar{c}}\psi^c\bar{\psi}\bar{b}\Big). \end{aligned} \tag{54}$$

3) After canonical quantisation with normal ordering we get the quantum supercharges

$$\begin{cases} \hat{Q} = -ie^m_c\psi^c\Big(\partial_m + \omega_{a\bar{b},m}\psi^a\frac{\partial}{\partial\psi^b}\Big) \\ \hat{Q}^\dagger = ie^{\bar{m}}_{\bar{c}}\frac{\partial}{\partial\psi^{\bar{c}}}\Big(\partial_{\bar{m}} + \frac{1}{2}\ln(\det h) + \omega_{a\bar{b},\bar{m}}\psi^a\frac{\partial}{\partial\psi^b}\Big) \end{cases} \tag{55}$$

The second supercharge is obtained by the conjugation of $\hat{Q}$.

1.2.3 Outcomes

By a similar analysis one can determine the action of $\hat{Q}$ on a "holomorphic"wave function with fermionic charge p:

$$\hat{Q}A_{m_1\ldots m_p}(z,\bar{z})\psi^{m_1}\ldots\psi^{m_p} = -i\partial_m A_{m_1\ldots m_p}\psi^m\psi^{m_1}\ldots\psi^{m_p} \tag{56}$$

This is exactly fow ∂ acts on the forms from the Dolbeault complex if we substitute ψ^k by dz^k. Our SQM is dual to the following complex on M:

$$\Omega^{(0,0)} \to \Omega^{(0,1)} \to \Omega^{(0,2)} \to \ldots \tag{57}$$

Cohomology classes of that complex correspond to the vacuums of our SQM. The adjoint $\partial^\dagger$ is dual to $\hat{Q}^\dagger$.

2 $N = 4$ SUSY sigma-models

2.1 Kahler manifolds

Let (M,g) be an hermitian manifold. The Kahler form of g is a 2-form defined as

$$\Omega = g(J\_, \_), \tag{58}$$

where J is the complex structure tensor. M is called Kahler if $d\Omega = 0$. We list here some useful properties of Kahler manifolds:
1) Hermitian matrix from (17) satisfies:

$$\partial_n h_{m\bar{p}} - \partial_m hn\bar{p} = 0 \tag{59}$$

2) Locally h admits a following description:

$$h_{m\bar{n}} = \partial_m\partial_{\bar{n}}K(z,\bar{z}), \tag{60}$$

K is called the Kahler potential.
3) The criteria for M being Kahler:

$$\nabla_P J = 0 \tag{61}$$

4) The only non-zero Christoffel symbols are

$$\Gamma^p_{mn} = h^{\bar{q}p}\partial_m h_{n\bar{q}}, \ \ \Gamma^{\bar{p}}_{\bar{m}\bar{n}} = h^{\bar{p}q}\partial_{\bar{m}}h_{q\bar{n}} \tag{62}$$

We require the sigma-model from section 1.1.1 on an even-dimensional real M to be invariant under the extended $N=4$ SUSY algebra and will prove that this is possible if and only id M is Kahler. This will lead to an interesting result about Kahler manifolds:
1) For Kaher manifolds the Dolbeault and anti-Dolbeault laplacians are equal:

$$\{\partial, \partial^\dagger\} = \{\bar{\partial}, \bar{\partial}^\dagger\} \tag{63}$$

2) $\{\partial, \bar{\partial}^\dagger\} = \{\partial^\dagger, \bar{\partial}\} = 0$

2.2

Now let's consider the sigma-model built from $2d$ real superfields (3):

$$S = \frac{1}{2}\int dt\; d\bar{\theta}\; d\theta g_{MN}\bar{D}X^M DX^N \tag{64}$$

To get 4 supercharges instead of 2, let's require the invariance of the action under the extended algebra (?):

$$\delta X^M = I^M_N(X)\Big(\epsilon DX^N - \bar{\epsilon}\bar{D}X^N\Big) \tag{65}$$

One might prove the following:
The action (32) admits a (33) symmetry iff I^N_M is an integrable almost complex structire and $\nabla_P I = 0$, i.e M is complex and Kahler. The condition $I^2 = -id$ and integrability is necessary and sufficient for the transformations to form the necessary SUSY algebra, the covariant derivative of I being 0 is equivalent to the action being invariant under this algebra.
Therefore, in the case of M being Kahler we get 4 supercharges that can be quantised with a well-known procedure. Let's just write them out:

$$\begin{cases} \hat{Q}_0 = -\frac{i}{\sqrt{2}}\Big(\psi^m D_m + \psi^{\bar{m}} D_{\bar{m}}\Big) \\ \hat{Q}_0^\dagger = -\frac{i}{\sqrt{2}}\Big(e^m_{\bar{c}}\frac{\partial}{\partial\psi^{\bar{c}}} D_m + e^{\bar{m}}_{\bar{c}}\frac{\partial}{\partial\psi^{\bar{c}}} D_{\bar{m}}\Big) \\ \hat{Q}_1 = \frac{1}{\sqrt{2}}\Big(\psi^{\bar{m}} D_{\bar{m}} - \psi^m D_m\Big) \\ \hat{Q}_1^\dagger = \frac{1}{\sqrt{2}}\Big(e^{\bar{m}}_c\frac{\partial}{\partial\psi^c} D_{\bar{m}} - e^m_c\frac{\partial}{\partial\psi^c} D_m\Big), \end{cases} \tag{66}$$

where $D_m = \partial_m + \omega_{a\bar{b}}\psi^a\frac{\partial}{\partial\psi^b}$. All of the above provides an elementray prood that the action of the operator $\psi^m D_m$ ($\psi^{\bar{m}}$) on the wave functions $\Psi(z,\bar{z},\psi)$ is dual to the action of the operator ∂ ($\bar{\partial}$) on the forms in the Kahler-de-Rham complex. Thus, $\hat{Q}_0$ is dual to the $\partial + \bar{\partial}$, $\hat{Q}_1$ is dual to the $\bar{\partial} - \partial$. By construction, the supercharges, and therefore, the operators acting on forms satisfy the $N=4$ SUSY algebra:

$$\begin{cases} \{\hat{Q}_i, \hat{Q}_j\} = \{\hat{Q}_i^\dagger, \hat{Q}_j^\dagger\} = 0 \\ \{\hat{Q}_i, \hat{Q}_j^\dagger\} = \delta_{ij}\hat{H} \end{cases} \tag{67}$$

from which the previously announced theorem follows.

2.3 Lefshetz decomposition and R-symmetry